

Homework 40 Solution: The Geometry of Vector Spaces

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BCS All Sections

Problem 1. Affine Combination. Write $\mathbf{y}$ as affine combination of $\mathbf{v}_1$, $\mathbf{v}_2$, and $\mathbf{v}_3$.

$$\mathbf{v}_1 = \begin{bmatrix} -3 \\ 1 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 0 \\ -4 \\ 2 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 4 \\ -2 \\ 6 \end{bmatrix}, \quad \mathbf{y} = \begin{bmatrix} 17 \\ 1 \\ 5 \end{bmatrix}.$$

Solution. Method I: Using Homogenous Vectors

By using homogenous forms $\tilde{\mathbf{v}}_1, \tilde{\mathbf{v}}_2, \tilde{\mathbf{v}}_3$, and $\tilde{\mathbf{y}}$, one can write:

$$\begin{bmatrix} -3 & 0 & 4 & 17 \\ 1 & -4 & -2 & 1 \\ 1 & 2 & 6 & 5 \\ 1 & 1 & 1 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 4 & -2 & 1 \\ 1 & -2 & 6 & 5 \\ -3 & 0 & 4 & 17 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 3 & -3 & 0 \\ 0 & -3 & 5 & 4 \\ 0 & 3 & 7 & 20 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 10 & 20 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 1 & -1 & 0 \\ 0 & 0 & 1 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Implies

$$\mathbf{y} = -3\mathbf{v}_1 + 2\mathbf{v}_2 + 2\mathbf{v}_3.$$

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Problem 2. Flat – A Universe Without the Zero. Explain why $x + 2y + 7z = 1$ is a Flat. Furthermore, write the dimension of the Flat.

Solution. The given equation is $x + 2y + 7z = 1$. One can write

$$x = 1 - 2y - 7z.$$

Implies

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 - 2y - 7z \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \begin{bmatrix} -2y \\ y \\ 0 \end{bmatrix} + \begin{bmatrix} -7z \\ 0 \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix} + \underbrace{\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix} y + \begin{bmatrix} -7 \\ 0 \\ 1 \end{bmatrix} z}_{\text{Span}\left\{\begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} -7 \\ 0 \\ 1 \end{bmatrix}\right\} \cong \mathbb{R}^2, \text{ a plane}}.$$

Let $y = t, z = s$, and

$$\mathbf{w} = \begin{bmatrix} x \\ y \\ z \end{bmatrix}, \quad \mathbf{p} = \begin{bmatrix} 1 \\ 0 \\ 0 \end{bmatrix}, \quad \mathbf{u} = \begin{bmatrix} -2 \\ 1 \\ 0 \end{bmatrix}, \quad \mathbf{v} = \begin{bmatrix} -7 \\ 0 \\ 1 \end{bmatrix}.$$

Implies

$$\mathbf{w} = \mathbf{p} + t\mathbf{u} + s\mathbf{v}.$$

Hence, $x + 2y + 7z = 1$ is a flat passing through $\mathbf{p}$ parallel to a plane spanned by $\mathbf{u}$ and $\mathbf{v}$. The flat is two dimensional.

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Problem 3. Affine Independence. Show that the set S is affinely independent.

$$S = \left\{ \begin{bmatrix} 1 \\ -1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ -2 \\ 0 \end{bmatrix} \right\}.$$

Solution.

Let

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ -1 \\ 2 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 2 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 1 \\ 2 \\ -2 \\ 0 \end{bmatrix}.$$

Now

$$\mathbf{v}_2 - \mathbf{v}_1 = \begin{bmatrix} 1 \\ 2 \\ -2 \\ 0 \end{bmatrix}, \quad \mathbf{v}_3 - \mathbf{v}_1 = \begin{bmatrix} 0 \\ 3 \\ -4 \\ -1 \end{bmatrix}.$$

Implies

$$\alpha_1(\mathbf{v}_2 - \mathbf{v}_1) + \alpha_2(\mathbf{v}_3 - \mathbf{v}_1) = \mathbf{0}.$$

So

$$\begin{bmatrix} 1 & 0 & 0 \\ 2 & 3 & 0 \\ -2 & -4 & 0 \\ 0 & -1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 3 & 0 \\ 0 & -4 & 0 \\ 0 & -1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & -4 & 0 \\ 0 & -1 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

Implies

$$\alpha_1 = \alpha_2 = 0.$$

Implies $\mathbf{v}_2 - \mathbf{v}_1$ and $\mathbf{v}_3 - \mathbf{v}_1$ are linearly independent. Henceforth, $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$ is affinely independent.

Problem 4. Barycentric Coordinates. Find the Barycentric coordinates of p with respect to the affinely independent set S of points that precedes it.

$$S = \left\{ \begin{bmatrix} 1 \\ -1 \\ 2 \\ 1 \end{bmatrix}, \begin{bmatrix} 2 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 2 \\ -2 \\ 0 \end{bmatrix} \right\}, \quad p = \begin{bmatrix} 5 \\ 4 \\ -2 \\ 2 \end{bmatrix}.$$

Solution.

$$\begin{bmatrix} 1 & 2 & 1 & 5 \\ -1 & 1 & 2 & 4 \\ 2 & 0 & -2 & -2 \\ 1 & 1 & 0 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 & 5 \\ 0 & 3 & 3 & 9 \\ 0 & -4 & -4 & -12 \\ 0 & -1 & -1 & -3 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 & 5 \\ 0 & 1 & 1 & 3 \\ 0 & 1 & 1 & 3 \\ 0 & 1 & 1 & 3 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 & 5 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 2 & 1 & 5 \\ 0 & 1 & 1 & 3 \\ 0 & 0 & 1 & -1 \\ 0 & 0 & 0 & 0 \end{bmatrix}.$$

Barycentric coordinates are

$$\begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} -2 \\ 4 \\ -1 \end{bmatrix}.$$

Problem 5. Convex Combination. Let

$$\mathbf{v}_1 = \begin{bmatrix} 1 \\ -1 \\ 2 \\ 1 \end{bmatrix}, \quad \mathbf{v}_2 = \begin{bmatrix} 1 \\ -1 \\ 2 \\ 1 \end{bmatrix}, \quad \mathbf{v}_3 = \begin{bmatrix} 1 \\ -1 \\ 2 \\ 1 \end{bmatrix},$$

and

$$\mathbf{p}_1 = \begin{bmatrix} 1 \\ -1 \\ 2 \\ 1 \end{bmatrix}, \quad \mathbf{p}_2 = \begin{bmatrix} 1 \\ -1 \\ 2 \\ 1 \end{bmatrix}, \quad \mathbf{p}_3 = \begin{bmatrix} 1 \\ -1 \\ 2 \\ 1 \end{bmatrix}, \quad \mathbf{p}_4 = \begin{bmatrix} 1 \\ -1 \\ 2 \\ 1 \end{bmatrix}.$$

Let S be the orthogonal set $\{\mathbf{v}_1, \mathbf{v}_2, \mathbf{v}_3\}$. Determine whether each $\mathbf{p}_i$ is in $\text{Span } S$, $\text{aff } S$, and $\text{conv } S$.

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Good Luck

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